Drop It — Investigation

Chapter 3 · Use with Lesson 4 (Day 2)

Name: _____ Date: _____ Group: _____

BIG QUESTION: How does changing the drop height affect the change caused by the same falling object?

PREDICT

At which height will the ball make the biggest crater? Why?

FAIR TEST

We change: drop height (30 cm · 60 cm · optional 90 cm)

We keep the same: ball · sand tray · surface (re-smoothed) · drop technique

We measure: crater width in cm (depth is just shallow / medium / deep)

DATA TABLE

Run 3 trials at each height. Measure crater width in cm. Then find the average.

Drop Height	Trial 1 (cm)	Trial 2 (cm)	Trial 3 (cm)	Average Crater Width (cm)
30 cm				
60 cm				
90 cm (optional)				

OBSERVATION

Which height caused the greatest observable change? (Also note the crater depth: shallow / medium / deep.)

EXPLANATION

How does this evidence connect to energy of motion? (Remember: you measured the crater, not speed or energy.)
